Ethan Liu

Frontend Developer

155-1027-5677 · lhw1953927853@gmail.com **Available to start immediately** · Open to **full-time internship** (**five days a week**) · **Three months or longer** internship · Accepting **nationwide** internships.

Education

B.S. in Computer Science at **Communication University of China**, Beijing, ChinaSep, 2023 – Jun, 2027 (expected graduation)
Minor in Mathematics, GPA 3.56/4.00

Work Experience

JD Inc., Beijing — Dec, 2025 – Apr, 2026

Related Projects

Personal Blog System — https://liuhongwei.org, Liu-Hong-Wei/Blog

Tech Stack: React, TypeScript, Vite, Tailwind CSS, Motion, React Router, Unified.js

- Built a frontend engineering system using React and Vite, utilizing TypeScript for component-level type constraints and high reusability. Unified ESLint and Prettier configurations to enhance code consistency.
- Optimized first-screen performance by implementing route lazy loading and component splitting to reduce initial bundle size. Integrated Web Vitals monitoring for P75 analysis to drive critical resource loading optimization, stabilizing core page LCP under 2.5s.
- Encapsulated the data request layer with React Suspense and custom caching strategies to minimize duplicate requests and reduce redundant network overhead during cross-post navigation.
- Developed a robust Markdown rendering engine based on the Unified ecosystem, supporting header anchor generation, syntax highlighting, and extensions, while combining custom Hooks and Motion for enhanced interactive animations.
- Configured optimal Vite build strategies by splitting third-party dependencies and business modules using `manualChunks`, improving cache granularity and long-term hit rates.

Low-Code AI Workflow Building Platform — Liu-Hong-Wei/aiflow-studio

Tech Stack: React, TypeScript, React Flow, Zustand, NestJS, React Router

- Built a visual workflow orchestration system using **React Flow**, designing a DAG structure and execution DSL for nodes and edges, combining **Zustand** for unified state management and incremental updates to resolve data consistency and rendering performance issues under complex dependencies.
- Optimized streaming execution and long-session rendering by implementing real-time output and state feedback with **SSE**, and integrated `react-virtuoso` virtual lists in the chat area to improve rendering performance and user experience in long-message scenarios.
- Implemented a workflow execution engine based on **NestJS**, parsing orchestration results into DAG execution plans, designing node scheduling and task execution models that support parallel execution, failure retries, and interruption recovery.
- Abstracted the **Agent execution mechanism** by designing a Plan-Act-Observe inference flow, unifying LLM inference and Tool invocation into standard node semantics, and achieving capacity decoupling and dynamic extension based on a Tool Registry.
- Modularized **RAG capabilities**, completing document parsing, chunking, vector storage, and multi-way recall strategies, integrating them as standard nodes to participate in workflow orchestration and execution alongside the inference pipeline.

Skills

- Familiar with HTML5, CSS3, and JavaScript, and have experience with TypeScript.
- Familiar with React, Redux (including Redux Toolkit), React Router, and state management solutions.
- Familiar with build tools like Vite, Webpack, and package managers such as npm or pnpm, and support code splitting and performance optimization.
- Proficient in using Git for version control and collaborating on projects through platforms like GitHub/GitLab.
- Familiar with using AI programming tools like GitHub Copilot, and have experience integrating AI capabilities into development workflows.